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Key Points:

- We incorporated the Model for Simulating Aerosol Interactions and Chemistry aerosol module in Community Earth System Model version 2 (Community Atmosphere Model Version 6 with interactive chemistry) model
- We found small radiative forcing due to nitrate-radiation interactions but large radiative forcing due to nitrate-cloud interactions
- We found rapid increases in nitrate aerosol REari in China and India since 1975 and positive nitrate aerosol REaci in China after 2000

Supporting Information:

Supporting Information may be found in the online version of this article.

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Radiative Forcing of Nitrate Aerosols From 1975 to 2010 as Simulated by MOSAIC Module in CESM2-MAM4

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Abstract We incorporate the Model for Simulating Aerosol Interactions and Chemistry (MOSAIC) module in the Community Earth System Model version 2 (CESM2) Community Atmosphere Model Version 6 with interactive chemistry (CAM6-chem), and couple it with the four mode version of the Modal Aerosol Module (MAM4). The MOSAIC module is used to simulate the thermodynamics of the gas-aerosol mass exchange, with a special focus on simulating nitrate aerosol. By comparing against ground and satellite observations, we found that the MOSAIC/MAM4 scheme performs reasonably well in simulating spatiotemporal distributions of aerosols, including nitrate aerosol. We conducted a series of model experiments with and without nitrate aerosols, and examined the radiative effect (RE) associated with nitrate aerosols in 1975, 2000, and 2010, and accessed the radiative forcing (RF) of nitrate aerosols between the present day and pre-industrial periods. Comparing with the nitrate aerosol RE, we predicted relatively small RF of anthropogenic nitrate aerosol from aerosol-radiation interactions (RF_{ari}: -0.014 W m^{-2}) and large RF from aerosol-cloud interactions (RF_{aci}: -0.219 W m^{-2}). Regional signatures of nitrate RE/RF are noticeable and important: for instance, very small changes in RE_{ari} in Europe and USA, but 2.8–3 times increases in RE_{ari} in India and China from 1975 to 2010, while RE_{aci}/RF_{aci} in China is a warming effect due to the competing effect between sulfate and nitrate aerosols as cloud condensation nuclei.

1. Introduction

Nitrate is one of the most important atmospheric aerosols that greatly impacts global nitrogen cycles, climate, and human health (Myhre, Shindell, et al., 2013, IPCC AR5 Chapter 8; Pagalan et al., 2018). Nitrate aerosols can affect climate through scattering shortwave (SW) radiation and acting as cloud condensation nuclei (CCN) affecting cloud microphysical properties. Anthropogenic nitrate aerosols are usually formed from gas precursors such as nitrogen oxides (NO_x), which is largely related to heavy air pollution caused by economic activities. Aerosol mass spectrometer measurements over the globe show that nitrate aerosols can contribute a significant fraction to total aerosol mass, especially over metropolitan areas during winter according to the measurements (Zhang et al., 2007).

Despite the importance of nitrate aerosols, only a handful climate models simulate their formation and lifecycle in the atmosphere (Aan de Brugh et al., 2011; Adams et al., 2001; An et al., 2019; Bellouin et al., 2011; Bian et al., 2017; Feng & Penner, 2007; Hauglustaine et al., 2014; Lamarque et al., 2012; Li et al., 2015; Liao et al., 2004; Myhre, Samset, et al., 2013; Rodriguez and Dabdub, 2004; Skeie et al., 2011; Xu & Penner, 2012), not even for many of the models that participated in the Coupled Model Intercomparison Project 5 (CMIP5). To simulate the gas-aerosol exchange, Adams et al. (2001) and Liao et al. (2004) used the aerosol thermodynamic equilibrium model ISORROPIA (Nenes et al., 1998), which considered different aerosol types such as ammonium, sulfate, nitrate and sea salt (as sodium and chloride). In the aerosol module of Liao et al. (2004), the soluble part of dust is not considered, but the dust particles are allowed to uptake sulfur dioxide (SO_2) and nitric acid gas (HNO_3) on their surface. Bian et al. (2017) examined the nitrate aerosols simulated by nine global models, which participated in the AeroCom Phase III study. They found that atmospheric burdens of HNO_3 differ by a factor of nine, although the production of HNO_3 is treated in similar approaches (i.e., all considered O_3-NO_x-HO_x chemistry and N_2O_5 hydrolysis). All nine models use the thermodynamic equilibrium model (TEQM) to simulate the gas-aerosol partitioning of nitrate aerosols, but still differ in many aspects, for instance, with differences in (a) how aerosol size and composition affect

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gas-aerosol partitioning, (b) whether nitrate aerosols can form on sea salt and dust particles; and (c) computational configurations/approaches. As a result, global atmospheric burdens of nitrate aerosols differ by a factor of 13 among these models. Instead of using TEQM, Feng and Penner (2007) adopted a hybrid dynamical approach (HDYN), which assumes that the condensation of HNO_3 on pre-existing aerosols occurs in a non-equilibrium state. It was found that the HDYN approach predicts less nitrate aerosols compared to the TEQM approach.

The Model for Simulating Aerosol Interactions and Chemistry (MOSAIC) (Zaveri et al., 2008) has been implemented in the Weather Research and Forecasting Model with Chemistry (WRF-Chem) and used in many regional studies (Fast et al., 2006; Yang et al., 2011). The MOSAIC is a comprehensive aerosol chemistry module designed to dynamically partition H_2SO_4, MSA, HNO_3, HCl, NH_3 and secondary organic aerosol (SOA) to different aerosol size bins/modes, driven by particle-phase thermodynamics and heterogeneous chemical reactions. Several novel methods (e.g., "dynamic pH" and "adaptive time stepping") are used in MOSAIC to ensure accurate yet computationally efficient solutions. The first implementation of MOSAIC in a global climate and chemistry model was in the Community Atmosphere Model version 5, the atmosphere component, with chemistry (CAM5-chem), of the Community Earth System Model version 1 (CESM1) and coupled with the 7-mode version of the Modal Aerosol Module (MAM7) (Liu et al., 2012) and MOZART gas-phase chemistry (Zaveri et al., 2021). In this study, we further implement the MOSAIC module in the 4-mode version of the Modal Aerosol Module (MAM4) (Liu et al., 2016) in the CAM version 6 with chemistry (CAM6-chem). The CAM6 is the atmosphere component of the CESM version 2 (CESM2).

Many modeling studies have demonstrated the weakening of projected future aerosol radiative forcing; however, nitrate aerosol is an exception (Bauer et al., 2007; Hauglustaine et al., 2014; Myhre, Shindell, et al., 2013). Due to the projected reduction in SO_2 emissions in the future, nitrate aerosols will contribute more to the total atmospheric aerosol mass, and hence cast relatively more significant radiative forcing (Bauer et al., 2007; Bellouin et al., 2011; Li et al., 2015). Liao et al. (2004) used the Goddard Institute Space Study (GISS) II incorporated with ISORROPIA and estimated the radiative forcing due to nitrate aerosol-radiation interaction is -0.14 W m^{-2} for year 1985 compared to year 1850. Based on several models' results, IPCC AR5 estimated the radiative forcing due to nitrate aerosol-radiation interaction to be -0.11 W m^{-2}, with an uncertain range from -0.30 to -0.03 W m^{-2}. In Myhre, Samset, et al. (2013), this range is from -0.12 to -0.02 W m^{-2} according to the AeroCom Phase II models prediction.

Bellouin et al. (2011) estimated that the direct and first indirect aerosol forcing associated with nitrate aerosols increases from -0.17 W m^{-2} to -0.36 W m^{-2} in terms of magnitude from year 2000 to year 2090 based on the Representative Concentration Pathway 4.5 (RCP4.5). These estimates of radiative forcing associated with nitrate aerosols are important; however, we note that, in these studies, radiative forcing due to nitrate aerosol-cloud interaction (RF_{aci}) is not fully represented, and the radiative forcing due to nitrate aerosol-radiation interaction (RF_{ari}) estimated by different models has large uncertainty.

In a retrospective point of view, the radiative forcing of nitrate aerosols varies from region to region, as well as from time to time, since it is greatly associated with human economic activities (i.e., NO_x and NH_3 emissions). Some developing countries such as China and India have experienced severe air pollution because of heavy traffic and industrial activities since 1980s (Lu et al., 2011). In recent years, China and India together contribute about 30%–40% of global emissions of some aerosol species and aerosol precursors (e.g., SO_2, black carbon (BC), organic carbon, and NO_x) (Hoesly et al., 2018; Smith et al., 2011). Besides, coming with significant health impacts (Pöschl, 2005), these aerosol pollutants can alter local weather/climate through aerosol-radiation and aerosol-cloud interactions, which in turn affect, and most of the time exacerbate, aerosol concentrations (Gao et al., 2011; Huang et al., 2020; Zhao et al., 2006). On the contrary, in developed countries/regions, declining trends in NO_x emissions have occurred due to strict regulations. How the radiative effects of nitrate vary in different regions in the last 40 years is the focus of this study.

This paper is organized as follows: in Section 2, we briefly describe the CESM2 model and its aerosol component MAM4; then we introduce the model configurations and observational datasets used in this study. In Section 3, we validate modeled nitrate aerosols against some observational data and examine changes in other aerosol types caused by nitrate aerosols. In addition, we assess the radiative effects associated with

10.1029/2021JD034809

nitrate aerosols in 1975, 2000, and 2010 and examined the radiative forcing of nitrate aerosols. Finally, we conclude the study in Section 4 with a discussion of future implications.

2. Methodology and Observation

2.1. Model Description

2.1.1. A Brief Description of CESM2.0

CESM2.0 is the recently released version 2 of the NCAR Community Earth System Model that can simulate the Earth's past, present, and future climate states (Danabasoglu et al., 2020). Compared to the previous version 1 of CESM (Hurrell et al., 2013), the atmosphere component of CESM2.0 is updated to the latest version 6 of Community Atmosphere Model (Bogenschutz et al., 2018), which contains an improved two-moment Morrison-Gettelman scheme for cloud microphysics (MG2) (Gettelman et al., 2015), and the Cloud Layers Unified by Binormals scheme (CLUBB) for boundary layer turbulence, shallow convection, and cloud macrophysics (Golaz et al., 2002; Larson et al., 2002). Compared to the previous version of cloud microphysics scheme (Morrison Gettelman, 2008), the new MG2 scheme is upgraded to include prognostic treatment of precipitation and new ice nucleation scheme (Shi et al., 2015; Wang et al., 2014), which leads to improved representation of mixed-phase and cirrus clouds. The 4-mode version of the Modal Aerosol Module (MAM4; Liu et al., 2016) is used to simulate the evolution of atmospheric aerosols, including processes of emissions, gas- and aqueous-phase chemistry, gas-aerosol exchange, nucleation, coagulation, and dry/wet depositions. MAM4 is fully coupled with cloud droplet activation and ice nucleation parameterizations (Abdul-Razzak & Ghan, 2000; Liu & Penner, 2005; Liu et al., 2007) as well as RRTMG radiation package (Iacono et al., 2008) so that both the aerosol-cloud and aerosol-radiation interactions are considered.

CESM2 is set to run with the Community Emission Data System (CEDS) anthropogenic emissions Coupled Model Intercomparison Project 6 (CMIP6) (Hoesly et al., 2018), which use existing emission inventories, emission factors, and activity/driver data to estimate country-level and sector-specific emissions. Biomass burning emissions are those provided for CMIP6 (van Marle et al., 2017). Other emissions used in CESM2 are described in Emmons et al. (2020). To initiate CESM2 simulations, gridded reactive gases, aerosols, and aerosol and ozone precursor compounds with monthly seasonality for each model year are read in. For most aerosols and precursor gases, CEDS gives higher global mean emissions compared to CMIP5 emissions since 1970s as shown in Figure 2 in Hoesly et al. (2018). The sources of NO_x are important for our simulations. Besides the anthropogenic sources (including aircraft emissions), model also reads in emissions from biomass burning (equal to 17.7% of anthropogenic emissions) and soils (equal to 12.1% of anthropogenic emissions). In addition, NO emissions produced via lightning process are calculated online (equal to 5.3% of anthropogenic emissions). In total, the NO_x emissions from natural sources are about 35% of those from anthropogenic sources. The sources of NO_x and NH_3 emissions for different time periods used in this work and NO_x and NH_3 emissions used in previous studies are given in Table S1.

2.1.2. MOSAIC Module Coupled With MAM4

The default aerosol module in CESM2 is MAM4 (Liu et al., 2016). In MAM4, Aitken, accumulation, coarse, and primary carbon modes are considered. Considered types of aerosol species include sulfate, BC, primary particulate organic matter (POM), SOA, sea salt, and mineral dust. Aerosol particles are treated as internal mixture within the same mode, while aerosols in different modes are treated as an external mixture.

In this study, we couple the MOSAIC with MAM4 in CESM2. This requires that the host code (CAM6-chem + MAM4) simulate the relevant aerosol and trace gas species used in MOSAIC, so several aerosol species are added or changed compared to the default MAM4. First, nitrate aerosol is added. Currently, most of the physical properties (including radiative properties) of nitrate aerosol are set to those of sulfate, except for its density, molecular weight, and hygroscopicity. The value of nitrate hygroscopicity, 0.67, is adopted from (Petters & Kreidenweis, 2007) and used in studies like Xu and Penner (2012). Second, ammonium aerosol (NH_4^+) is explicitly treated so that sulfate and nitrate can compete for positive ions in the particle phase, and the sulfate species is pure sulfate versus ammonium bisulfate in the default MAM4. In the default version of MAM4, primary sea salt aerosol is a single bulk/lumped aerosol tracer. With MOSAIC, this is replaced by explicit sodium (Na^+) and chloride (Cl^-) species, and primary sea-salt emissions are 33.5% Na^+, 53.8%

Table 1
Newly Added Aerosol Species (O) in Four Aerosol Modes

Species	Accumulation (a1)	Aitken (a2)	Coarse (a3)	Primary carbon (a4)
BC	X			X
POM	X			X
SOA	X	X		
SO_4	X	X	X	
NH_4	O	O	O	
NO_3	O	O	O	
Cl	X	X	X	
Na	O	O	O	
Dust	X	X	X	
Ca	O	O	O	
CO_3	O	O	O	
Total	11	9	8	2

Cl^-, and 7.7% of SO_4^{2-} by mass. Also, mineral dust is now a mixture of 95% other inorganic (OIN), 2% calcium (Ca^{2+}) and 3% carbonate (CO_3^{2-}) by mass. This is a difference compared to ISORROPIA used in Adams et al. (2001) and Liao et al. (2004), since the influence of carbonate calcium on the aerosol thermodynamics is considered in MOSAIC. Except for molecular weights, the physical properties of the Na^+ and Cl^- species are set to those of the default bulk sea salt, and the physical properties of the OIN, Ca^{2+}, and CO_3^{2-} are set to those of the default bulk soil dust. As shown in Table 1, a total of 15 new tracers are added to the MAM4 module. Note that nitrate, ammonium, and chloride are allowed to form on the primary carbon mode, but like sulfate are immediately transferred to the accumulation mode through the primary carbon aging treatment.

The coupling involves replacing the default MAM4 treatments of trace gas condensation/evaporation to/from interstitial aerosol particles with the MOSAIC treatments. The MOSAIC treatment of SOA formation was changed to the default MAM4 treatment. In addition, the aqueous (i.e., cloud water) chemistry is modified to include the new species noted above.

MAM4/MOSAIC is coupled with the MOZART-TS1 (Model for Ozone and Related chemical Tracers, troposphere and stratosphere) chemistry mechanism used in CAM6-chem (Emmons et al., 2020). This is necessary because the formation of HNO_3 is through O_3-NO_x-HO_x chemistry or N_2O_5 hydrolysis, which are calculated in TS1 chemistry and not available in the CAM6 simplified chemistry. By adding these additional tracers, the computational expense of MOZART-MAM4-MOSAIC is around 15%–20% greater than that of MOZART-MAM4. The physical processes and chemical reactions related to nitrate aerosols in accumulation, Aitken, and coarse modes are listed in Table S2.

2.2. Model Experiment Setup

In this study, we run CESM v2.0.0 at 1-degree horizontal resolution. The model cases and configuration settings are listed in Table 2. Two sets of configurations are used for different purposes. For the purpose of model validation (Section 3.1), we select FCSD ("F" here stands for the CESM2 component set in which CAM and CLM (Community Land Model) are active but sea ice and ocean components are prescribed in the model. "C" here stands for MOZART-TS1 chemistry is turned on. "SD" here stand for specified dynamics) component set for CAM-chem with Specified Dynamics: http://www.cesm.ucar.edu/models/cesm2/config/compsets.html. The simulation is run for years 2005–2007, with 56 vertical layers and the meteorology nudged by MERRA2 re-analysis data. The simulation years are selected because of the data availability. This experiment is referred to as MSD (MOSAIC Specified Dynamics) case. In order to examine the aerosol radiative effects (Sections 3.4~3.5), we select FC2000climo ("F" and "C" here are the same as in FCSD, "2000climo" stands for climatology of 2000) component set, in which sea surface temperature (SST) and radiative agents other than aerosols/aerosol precursors (i.e., CO_2, CH_4) are fixed at year 2000 level. Here we design four groups of experiments, in which year 1850, 1975, 2000, and 2010 aerosol and aerosol precursors (e.g., SO_2 and NO_x) emissions are used. (Please note that meteorology is not nudged in these simulations). We run these four cases, referred to as M1850, M1975, M2000, and M2010 for 21 years as the first-year result is treated as spin-up and not included in the analysis. For the comparison purpose, aforementioned cases are accompanied with "noNO3" cases (i.e., MSD_noNO3, M1850_noNO3, M1975_noNO3, M2000_noNO3, and M2010_noNO3), in which the formation of nitrate aerosol is turned off while the MOSAIC aerosol module is still active (please note that these _noNO3 cases are not standard CAM-chem simulations since MOSAIC is used in gas-aerosol exchange calculations).

Table 2
Configuration of Model Experiments

CESM2 component set definition	Emission	Case name with NO_3	Case name without NO_3
FCSD (2005–2007)	CMIP6 (2005–2007)	MSD	MSD_noNO3
FC2000climo (21 years)	CMIP6 (1850)	M1850	M1850_noNO3
FC2000climo (21 years)	CMIP6 (1975)	M1975	M1975_noNO3
FC2000climo (21 years)	CMI6 (2000)	M2000	M2000_noNO3
FC2000climo (21 years)	CMIP6 (2010)	M2010	M2010_noNO3

10.1029/2021JD034809

By comparing the radiative fluxes modeled by cases with and without NO_3, we estimate the radiative effect (RE) of nitrate aerosols for different years, including both anthropogenic and natural sources. By contrasting REs between 2010 (present-day (PD) conditions) and 1850 (pre-industrial (PI) conditions), we estimate the radiative forcing (RF) of nitrate aerosols. To calculate RE_{aci} of nitrate aerosols, we use the methodology introduced in Ghan (2013). To facilitate the use of this method, without rerunning model simulations, we set up the simulations to calculate additional diagnostic radiative flux variables, in which the impact of either aerosols (clean condition) or clouds (clear condition) on radiation calculation are turned off. Therefore, the radiative effect due to aerosol-cloud interactions is calculated using following equation:

$$RE_{aci} = \Delta C_{clean} = \Delta\left(F_{clean} - F_{clean,clear}\right), \tag{1}$$

where Δ is an operator that represents the difference between the two cases with and without nitrate formation (e.g., M2000 – M2000_noNO3). C_{clean} is the diagnostic variable of cloud radiative forcing in an environment without aerosols' contribution on radiation. F is the net radiative flux at the top of the atmosphere (TOA), and F_{clean} and $F_{clean,clear}$ are the diagnostic variables of radiative flux at TOA with the contributions of all aerosols excluded and both aerosols and clouds excluded, respectively. Please note that, although the radiative effect is calculated using diagnostic variables, the aerosol-induced changes in cloud properties are calculated in the MG2 scheme.

To calculate RE of nitrate aerosols due to aerosol-radiation interactions (RE_{ari}) in M1850, M1975, M2000, and M2010 cases, we calculate another set of diagnostic radiative flux variables, in which only the contribution of nitrate aerosol to radiation is turned off. Then RE_{ari} of nitrate aerosols equals to the difference in net radiative fluxes at TOA between the default set and this diagnostic set for these four cases. (Please note that "noNO3" cases are not used in the RE_{ari} calculation.) The change in RE_{ari} of nitrate varies from 1975 to 2010 is compared against those of sulfate and BC aerosols, which are calculated using the same technique, to highlight the rapid increase of nitrate RE.

2.3. Observations

To evaluate modeled aerosol species, mainly nitrate, sulfate, and ammonium, we use observations from IMPROVE (Interagency Monitoring of Protected Visual Environments, http://vista.cira.colostate.edu/Improve) (Malm et al., 2004), CAWNET (China Atmosphere Watch Network) (Zhang et al., 2008), and EANET (Acid Deposition Monitoring Network in East Asia) networks. Totally 176 IMPROVE sites across the United States are available for comparison. The IMPROVE network uses the filter-type technique to measure fine particle masses, and speciated aerosol types every three days. In this study, we average records of sulfate and nitrate of each site into an annual mean and compare them against model results in the same grid box in scatter plots. Due to the small number of IMPROVE sites that monitor ammonium (only 13 sites over Southeast U.S.), comparison of modeled and observed ammonium over the United States is not included in this study. Using a similar filter-type technique, aerosol species mass concentrations at 29 EANET sites over East and Southeast Asia (namely Japan, South Korea, Russia, Mongolia, Thailand, Vietnam, Philippines and Indonesia) and at 14 CAWNET sites over China were measured. In this study, we validate model simulations of (a) annual mean surface mass concentration of aerosols against EANET and CAWNET observations, and (b) seasonality of surface mass concentrations of aerosols against CAWNET observations. We select observations of 2005 for IMPROVE, and select observations of 2006 for CAWNET and EANET, because of its availability.

The aerosol optical depth (AOD) modeled in the MSD case represents the total light extinction by aerosols in the vertical column. Accurate estimation of AOD is important for the calculation of RE. Therefore, we evaluate modeled AOD against NASA MODIS (Moderate Resolution Imaging Spectroradiometer) Level 3 collection 6 AOD products with 1-degree resolution (Levy et al., 2010) as well as NASA MISR (Multi-angle Imaging SpectroRadiometer) Level 3 collection 11 AOD products with 0.5-degree resolution (Kahn et al., 2007).

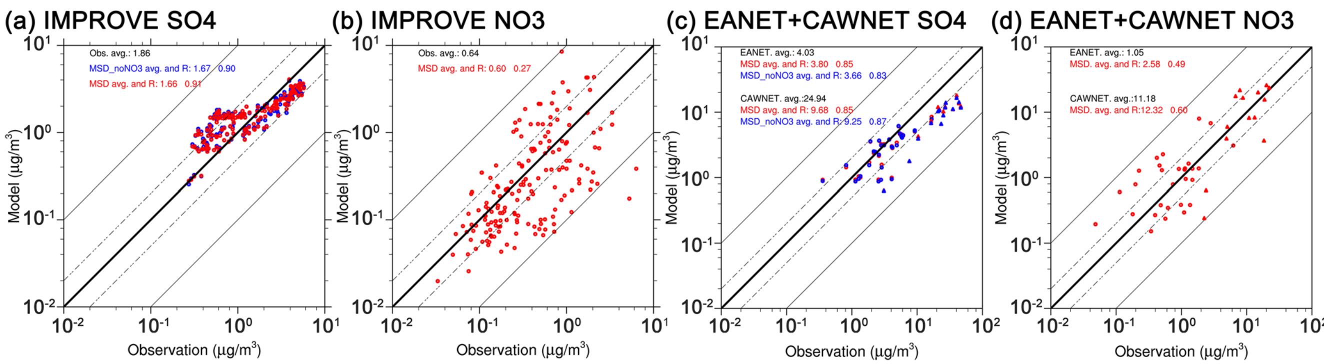

Figure 1. Scatter plots of modeled annual-average surface mass concentrations of sulfate (left panels) and nitrate (right panels) aerosols versus IMPROVE (upper panels, 2005) and CAWNET (triangle markers, 2006) + EANET (circle markers, 2006) (lower panels) observation networks. Note that only modeled and observed SO_4 and NO_3 mass in fine particles (with diameter <2.5 μm) are included in figures (a) and (b), respectively.

3. Results

3.1. Evaluation of Modeled Nitrate Aerosols With Observations

In Figure 1 and Table S3, we compare modeled surface concentrations of sulfate and nitrate aerosols against IMPROVE and CAWNET + EANET observations in scatter plots. IMPROVE networks only measure the mass of fine particles. Therefore, we also calculate the mass of modeled fine particles (with diameter <2.5 μm) and compare them against observations. The sulfate is reasonably simulated–the correlation coefficient for 176 sites is as high as 0.91 (MSD case). The MSD case slightly underestimates the annual mean averaged over sites compared to IMPROVE (1.67 μg/m^3 vs. 1.86 μg/m^3, normalized mean bias or NMB: −10.2%). Sulfate modeled by the MSD_noNO3 case performs slightly worse compared to the MSD case in terms of annual average. The model tends to overestimate sulfate by 100% (i.e., scatter points close to 1:2 ratio line) when sulfate surface concentrations are lower than 1 μg/m^3 but underestimates sulfate at higher concentrations. Most overestimations occur at the sites located in the western United States. As for the nitrate simulation, the correlation between modeled and observed values is much lower (0.27) compared to sulfate simulation; nevertheless, the correlation coefficient did pass 5% significance test. As shown in Figure 1b, scatter dots spread between 1:10 and 10:1 lines. On average, the model slightly underestimates nitrate surface concentration over the United States (0.6 μg/m^3 vs. 0.64 μg/m^3, NMB: −6.3%). It is interesting to note that two scatter dots are outside the 1:10 ratio line, indicating a significant underestimation of nitrate aerosols. Those two sites are located in heavily polluted cities in California, USA.

The surface mass concentrations of sulfate modeled by the MSD case is in a reasonable agreement with EANET observations (circle markers) in terms of annual average (3.80 μg/m^3 vs. 4.03 μg/m^3, NMB: −5.7%) and correlation (0.85), while the MSD_noNO3 case predicts about 3.7% less annual average value (3.66 μg/m^3). Similarly, MSD predicts slightly higher sulfate than MSD_noNO3 in China; however, both cases significantly underestimate the sulfate concentration by more than 50% compared to CAWNET (as shown by triangle markers): for example, many triangle markers align with 1:2 line in Figure 1c.

In Figure 2, we compare the seasonal variation of sulfate, nitrate, and ammonium modeled in the MSD and MSD_noNO3 cases against CAWNET observations. The results show that not only sulfate mass concentrations are significantly underestimated over China, but also the seasonality of sulfate mass concentrations in China is poorly simulated. For example, the mass concentration of sulfate at the surface peaked in winter as indicated by CAWNET observations shown in Figure 2a; However, both MSD and MSD_noNO3 cases predict much weaker seasonality with slightly higher concentrations during May and October. This is likely due to underestimation in SO_2 emissions but may also indicates a possibility that some chemical mechanisms related to sulfate formation are not well represented or missing especially in China in the model (Fan et al., 2018). One possible candidate is the quick sulfate formation in the neutral pH environment in China as discussed in Wang et al. (2016). This hypothesis deserves further investigation, but it is not in the scope of this study. By including the nitrate treatment (MSD vs. MSD_noNO3), the surface sulfate concentration is only slightly affected.

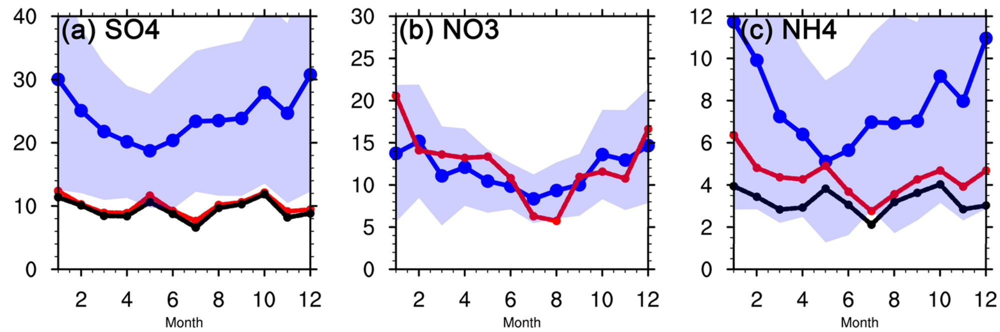

Figure 2. Seasonal cycles of surface mass concentration of key aerosol species, SO_4, NO_3, and NH_4 (unit: $\mu g\ m^{-3}$). Blue curves with blue shades represent CAWNET observation averaged over 14 stations as standard deviation. Red solid lines with dots indicate model simulation by MSD case, while black lines represent MSD_noNO3 case.

We found that the MSD case significantly overpredicts nitrate aerosol mass concentrations over the EANET sites by 145%. Meanwhile, surface mass concentrations of nitrate in China are well simulated in comparison to CAWNET, in terms of annual mean (Figure 1d, NMB: 10.2%) and seasonality (Figure 2b). The nitrate formation is slightly overestimated in the MSD case for January, March, and May but underestimated in July and August.

Figure 2c shows that the MSD case produces more ammonium aerosols than the MSD_noNO3 case does, which is due to the inclusion of nitrate aerosols. As a result, ammonium modeled in the MSD case is relatively closer to observations, especially during winter and spring but is still significantly underestimated. The large bias is due to the underestimation of sulfate by the model in the same period.

Figure 3 shows global AOD fields modeled in the MSD and MSD_noNO3 cases. The AOD fields in 2005 is compared against MODIS level 3 AOD products and MISR level 3 AOD products for the same time period. First, MODIS and MISR observations disagree with each other in some heavily polluted regions, such as China and India. AOD values retrieved from MODIS observations are apparently higher than from MISR observations. One noticeable example of the difference here is that MODIS AOD over China is quite high (>0.6) over the entire eastern China, but only a few regions (e.g., Jing-Jin-Ji region, Sichuan basin region, and Pearl River Delta region) have high AOD according to the MISR product. The disagreement between

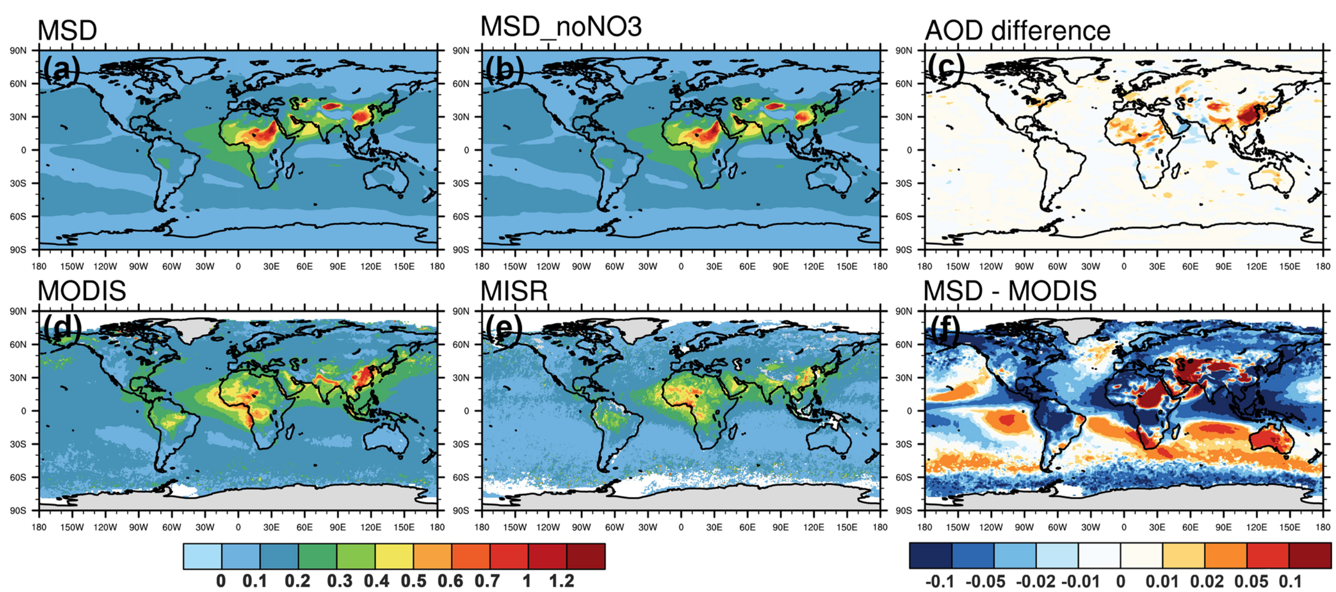

Figure 3. Modeled aerosol optical depth (AOD) fields for MSD and MSD_noNO3 cases and their difference in comparison with MODIS and MISR level 3 AOD products for the same period.

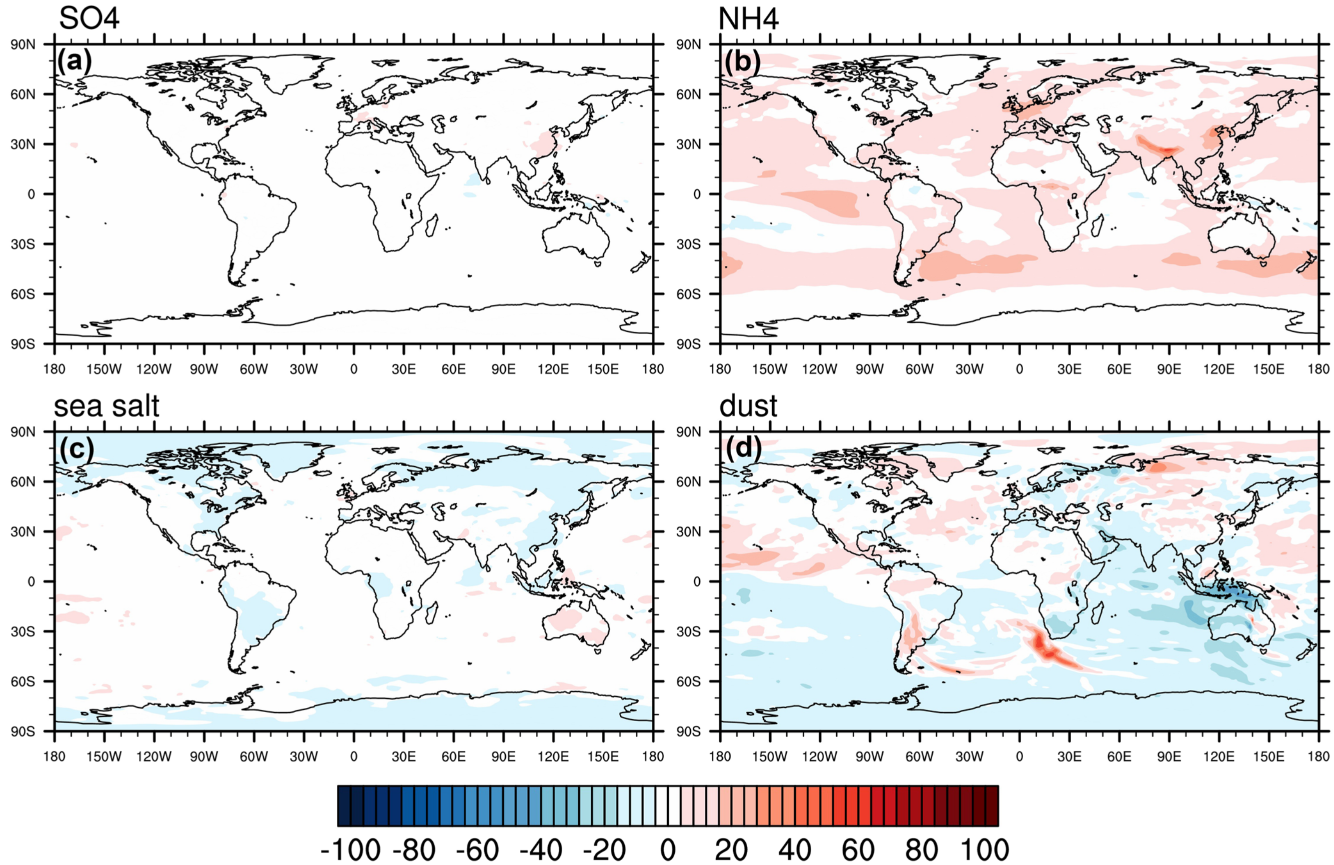

Figure 4. Relative differences in aerosol burdens (%) between MSD and MSD_noNO3 cases, calculated as 100× (MSD/MSD_noNO3-1).

MODIS AOD with MISR AOD is a well-known fact. Some studies (e.g., Kang et al., 2016) showed that MODIS retrieved AOD agrees better with ground-based AOD measurement AERONET than the MISR AOD product does.

In general, the model performs reasonably well. The MSD case successfully reproduces high AOD in central China but fails to capture high AOD in the northern China and southern coastal region of China as observed by both MODIS and MISR. MSD predicts AOD about 0.1 higher than MSD_noNO3 in many regions of China, a difference contributed by the treatment of nitrate aerosols, along with 20% more NH_4 as shown in Figure 4. Over the Indo-Gangetic plain, the AOD differences caused by adding nitrate aerosols and resulting additional 20–40% of NH_4 is around 0.02–0.05, while in eastern United States and Europe, AOD differences are around 0.01–0.02. Meanwhile, the AOD values around these regions are underestimated by the model compared to both MODIS and MISR observations by about 0.05, suggesting biases in other aerosol species. By examining two cases, we found that AOD near dust source regions are affected. For instance, higher AODs can be seen around Taklamakan Desert in China, Caspian Sea, Sahara Desert, and Australia. Higher AOD in Taklamakan Desert is partially due to higher dust emissions, while higher AODs in other regions are mostly due to nitrate aerosols.

Over ocean, AOD differences between MSD and MODIS range from −0.1 to 0.05. The model performs reasonably well over Northern Atlantic and Northeastern Pacific. The biases over other oceanic regions are caused by many possible factors, including biases in modeled outflow of anthropogenic and natural aerosol plumes as well as biases in retrieved MODIS AODs (e.g., over Southern Ocean >60ºS). Since this is not in the scope of this study, we will not further elaborate on the causes for modeled AOD bias. The seasonality of AOD simulations is discussed in the SI.

3.2. Changes in the Burdens of Other Aerosol Species

In Figure 4, we examine the percentage changes in modeled aerosol burdens (i.e., vertically integrated mass concentrations), including non-sea-salt sulfate (nss-sulfate), ammonium, primary sea salt (sum of sodium, sea salt chloride, and sea salt sulfate), and dust (sum of OIN dust, calcium, and carbonate) due to the introduction of nitrate by comparing the MSD and MSD_noNO3 cases. The inclusion of nitrate aerosols does not directly affect sulfate burden, since NH_3 will neutralize H_2SO_4 first then HNO_3, but indirectly through affecting the processes like wet removal. The results demonstrate that nss-sulfate aerosol across the globe is only slightly affected. In eastern China, the nss-sulfate burden is enhanced by less than 10%, while in the United States, Europe, and India, the nitrate treatment barely changes sulfate burden at all. The changes in nss-sulfate burdens occur mostly in the accumulation mode, simply because the accumulation mode sulfate accounts for 66% of the total sulfate mass. Ammonium is significantly affected, which is not surprising given the condition in which additional negative ion (NO_3^-) is introduced into the model. In Europe, India, and China, where nitrate aerosol burdens are high, we also see strong increases in ammonium, by as high as 40%–60%.

Primary sea salt aerosol burden slightly decreases because of the nitrate aerosols, especially over land. The reduction is because of uptake of HNO_3 gas by sea salt particles and resulting aqueous dissolution. Since sodium (Na^+) bonds with nitrate aerosols, we see a reduction in ss-chloride (Cl^-) burden and, therefore, the total sea salt burden as well (discussed later in Section 3.3).

The impact of nitrate aerosols on ammonium (positive) and sea salt aerosols (negative) is persistent regardless of other conditions (e.g., in the different years), which is not the case for dust aerosols. Taking the Central Asian dust field for instance, the dust burden decreases in MSD case but increases in the 1975 period (not shown). The different responses may be due to the climatic linkage of dust emissions (Xi & Sokolik, 2015) in different time periods. Nitrate aerosols can affect the dust burden through the following pathways: (a) condensed nitrate on dust surface can enhance its deposition; (b) nitrate's effect on climate can have a feedback on dust aerosol emission.

Comparing MSD and MSD_noNO3 cases, we find that the relative decreases in dust burden are very high over Australia because of enhanced precipitation (~0.5 mm d^{-1}, not shown) and resulting higher wet deposition rate in this region as shown in the dust budget analysis in Figure S1. The relative decreases in dust burden over Arabian Peninsula is also noticeable. This is because of reduced wind speed ($-0.1 \sim -0.2$ m s^{-1}, not shown) leading to decrease in dust emission shown in Figure S1. The relative change in dust burden over Taklamakan Desert in China is positive but small, which is due to enhanced dust emissions.

3.3. Nitrate Aerosols Budget Analysis

In Table 3, we summarize the budget of nitrate aerosols modeled by MSD case and compare the values against previous studies, including Xu and Penner (2012), Hauglustaine et al. (2014), and AeroCom phase III simulations (Bian et al., 2017). All the metrics for the MSD case are within the ranges predicted by the AeroCom Phase III models. The MSD case predicts a smaller burden but compared to the AeroCom study average and the other two studies. However, the MSD case predicts higher global-mean surface mass concentration of nitrate aerosols compared to the AeroCom study average, indicating that higher proportion of nitrate aerosol mass locates in lower altitudes in our simulation. (The vertical profiles of nitrate aerosols are discussed in the SI.) As shown in the Table 3, the lifetime of nitrate aerosol (3.3 days) in the MSD case is shorter than the AeroCom average and other two studies, due to relatively faster wet deposition rate. The total (all modes) net production rate is also somewhat lower in the MSD case. The lower production and shorter lifetime, which are due to numerous differences between the models, both contribute to the smaller burden in the MSD case.

The source terms for nitrate aerosols in Table 3 have somewhat different definitions for Xu and Penner (2012), Hauglustaine et al. (2014) and this study, due to different aerosol and chemical reaction treatments. The main chemical reactions for nitrate formation considered in these studies are: (a) the formation of NH_4NO_3 in aerosols, from gaseous phase HNO_3 and excess NH_3 after neutralizing sulfate; and (b) the heterogeneous reactions of HNO_3 on sea salt aerosols (e.g., formation of $NaNO_3$ when HNO_3 displacing chloride), and (c) the heterogeneous reactions of HNO_3 on dust aerosols (e.g., formation of $Ca(NO_3)_2$ when

Table 3
Nitrate Aerosol Budget for Years 2005–2007 (all Unit Converted to Tg-N)

	This study	AeroCom phase III. In Bian et al. (2017)	Xu & Penner, 2012	Hauglustaine et al., 2014
Burden (Tg-N)	0.11	[c] 0.14 [0.03, 0.42]	0.17	0.18
[a] Fine modes burden	27.3%	-	52.29%	27.8%
[b] Coarse modes burden	72.7%	-	47.71%	72.2%
Surface Concentration (μg-N kg^{-1})	0.09	0.05 [0.01, 0.11]	-	-
Dry Deposition (Tg-N a^{-1})	2.8	4.67 [0.25, 10.8]	3.96	1.66
Wet Deposition (Tg-N a^{-1})	9.3	10.4 [1.2, 20.5]	11.96	12.7
Total net production (Tg-N a^{-1})	12.3	[d] 15.07	15.9	14.4
Fine modes net production (Tg-N a^{-1})	[e] 5.14 (17.5, −36.8, 24.44)	-	8.5	3.2
Coarse modes net production (Tg-N a^{-1})	[e] 7.14 (6.7, −0.07, 0.51)	-	7.4	11.2
Lifetime (days)	3.3	5.0 [2.0, 7.8]	3.92	4.61

[a]The fine-mode burden in our study is the sum of Aitken mode and accumulation mode burdens. [b]The coarse-mode burden in (Hauglustaine et al., 2014) is the sum of dust and sea salt modes. [c]The values in brackets are the minimum and maximum values reported by AeroCom phase III models. [d]We assume that the total net production equals to the sum of wet and dry depositions in Bian et al. (2017). [e]Three items in parentheses are gas-aerosol exchange production, lost, and aqueous chemistry production in our study.

HNO_3 displaces carbonate). Note that the $CaCO_3$ content of dust particles varies among the studies (and Xu and Penner (2012) treat several other carbonates). In our study, we also treat HNO_3 uptake to cloud droplets, which are dominantly nucleated from the fine particle modes, and produces interstitial nitrate aerosols when cloud droplets evaporate. The NH_4NO_3 and heterogeneous reactions can occur in both fine and coarse mode aerosols in our study, but the fine particle nitrate is predominately NH_4NO_3 while the coarse particle nitrate is predominately $NaNO_3$ and $Ca(NO_3)_2$. In Hauglustaine et al. (2014), NH_4NO_3 is only formed in the accumulation mode (3.2 Tg-N a^{-1}), while the heterogenous reaction formations of nitrate aerosols only occur on the coarse mode dust and sea salt (11.2 Tg-N a^{-1}). In Xu and Penner (2012), the source terms for fine particle (<1.25 μm) and coarse particle (>1.25 μm) modes are listed separately because different numerical methods are used for these two modes of aerosols. The relative amounts of fine and coarse mode nitrate burdens and net production are comparable in our MSD case and Hauglustaine et al. (2014) while Xu and Penner (2012) found considerably more nitrate in the fine modes.

For our study, we also show the nitrate production and loss from gas-particle exchange and the net production from HNO_3 uptake to cloud droplets. For the fine modes, the gas-particle loss is quite large and exceeds the net production from HNO_3 uptake to cloud droplets in magnitude. This is due to the cycling of air parcels and aerosols into and out of clouds. When an air parcel moves into a cloud, many aerosol particles are activated to form cloud droplets, and most of the HNO_3 partitions to the cloud droplets. When an air parcel leaves a cloud, cloud droplets evaporate leaving liquid aerosol particles, and much of the nitrate they contain partitions back to HNO_3 gas.

In Figure 5, we examine the seasonality of nitrate aerosol burdens in three individual modes as modeled in the MSD case (including both interstitial and cloud-borne states). Nitrate mass in the Aitken mode is negligible compared to the other two modes. About 70% of the total nitrate mass is in the coarse mode, because of the formation of nitrate aerosols on sea salt and dust particles. Nitrate mass in the accumulation and coarse modes demonstrates a clear seasonality. The total mass burden of the coarse mode nitrate peaks in NH summer (JJA), because this is the season when the dust burden reaches the maximum (e.g., in outflow regions of Sahara Desert and Saudi Arabia).

As mentioned above, the accumulation mode nitrate aerosols are mostly formed from uptake of HNO_3 to cloud-borne aerosols and following the resuspension back to interstitial state. The solubility of HNO_3 is higher at lower temperatures (e.g., 15 times larger at 273 vs. 298 K) according to Henry's law. Therefore, in contrast to the coarse mode, the maximum total mass burden of the accumulation mode nitrate happens in winter (DJF), when the temperature in the northern hemisphere is the lowest and suitable for nitrate

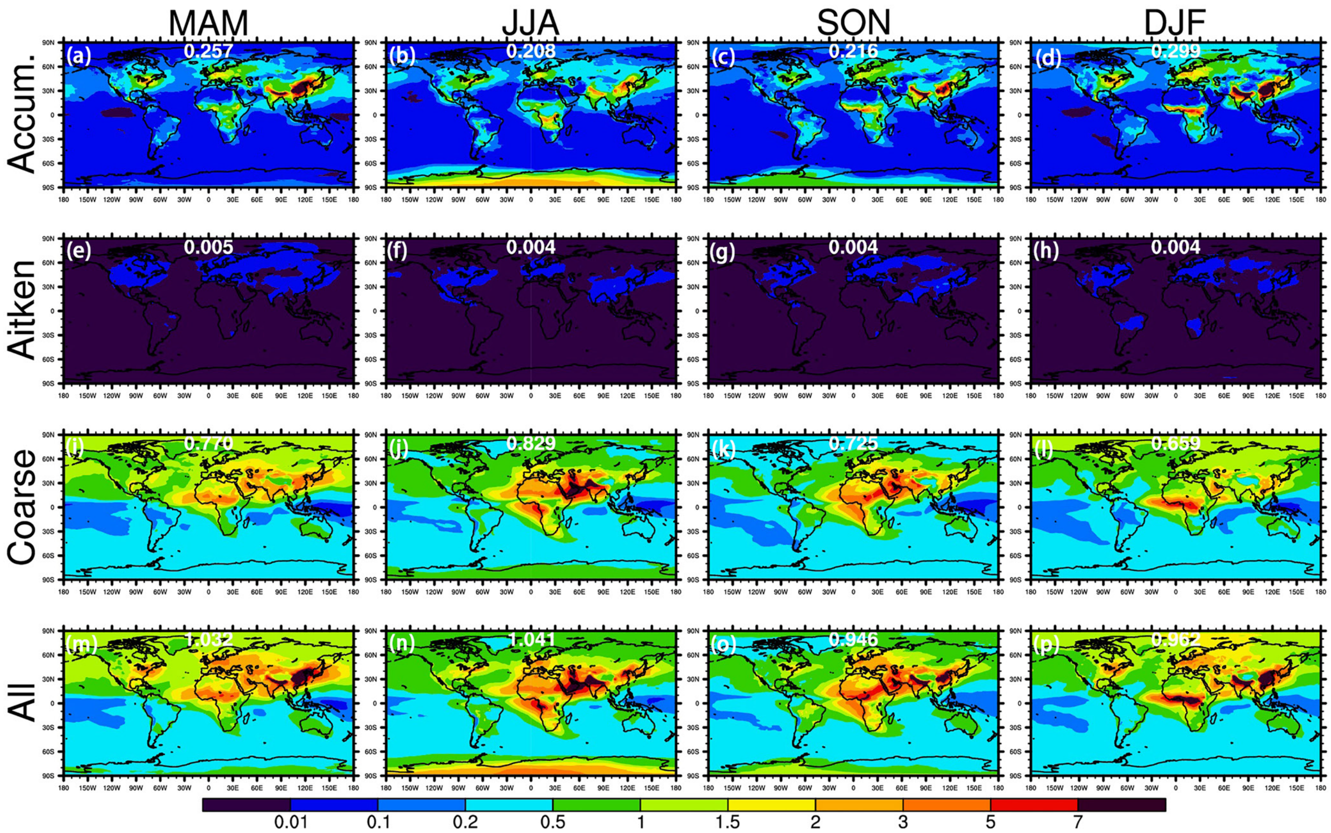

Figure 5. Seasonality of nitrate aerosol burdens (mg m^{-2}) in accumulation, Aitken, coarse, and all modes as modeled by MSD case.

formation. It is worth mentioning that, in heavily polluted regions, such as China and Indo-Gangetic plain, more nitrate mass is formed in the accumulation mode than in the coarse mode, especially when the temperature is relatively low. The nitrate burden in the accumulation mode can be as high as 12 mg m^{-2} in these regions. This is because: (a) accumulation mode nitrate is less volatile at lower temperatures, (b) the formation of coarse mode nitrate over land is calcium ion limited, and (c) a modest amount of calcium and sodium aerosols in the accumulation mode can promote the formation of nitrate aerosols from the excessive HNO_3.

Note that the seasonality of nitrate burden modeled by 7-mode version of MAM is discussed in the SI.

3.4. Radiative Effects of Nitrate Aerosol in Different Time Periods

In this section, we examine how the radiative effects of nitrate aerosols change from 1975 to 2010 using climatology simulations. In Figure 6, we plot the RE_{ari} of nitrate at TOA estimated from three experiments (i.e., M1975, M2000, and M2010). As shown in the bar charts of Figure 7, we further calculate the RF_{ari} averaged over the United States, Europe, China, and India in three time periods (only contiguous United States is considered in the calculation). Only eastern part of China (>105°E) is considered. Europe is defined as the European land in 35°N–70°N and 25°W–60°E.

As shown in Figure 6, the RE_{ari} fields are mostly determined by the accumulation mode of nitrate aerosols, because of higher aerosol number in this mode than in the coarse mode. In 1975, the globally averaged RE_{ari} of nitrate is −0.03 W m^{-2}. The cooling signal over ocean is generally below −0.1 W m^{-2}, and very small over the Southern Ocean, due to small nitrate burden in the accumulation mode. We can also find some cooling signals around −0.05 to −0.2 W m^{-2} over Africa, which is associated with the NO_x emissions from biomass burning aerosols. These patterns over ocean and Africa do not change much from 1975 to 2010. In 1975, the cooling effect of nitrate can be seen in large areas of Europe, the United States, eastern China, and

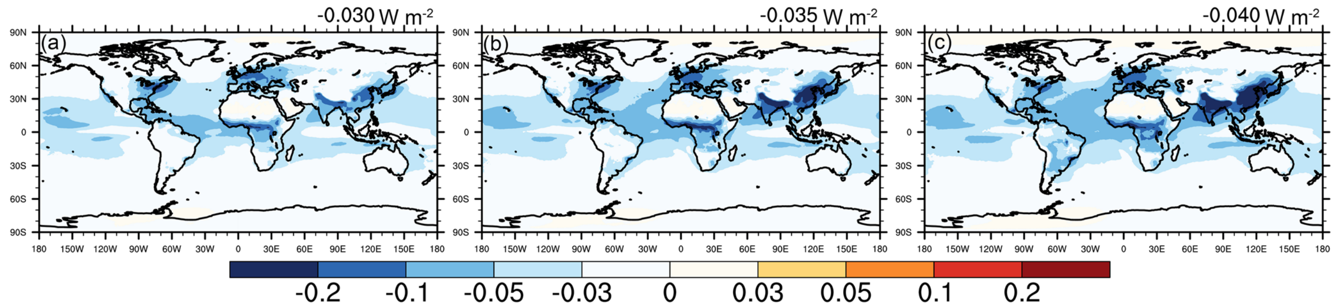

Figure 6. RE_{ari} of nitrate (unit: W m^{-2}) estimated from three pairs of experiments (from left to right: the difference in two simulations between M1975 and M1975_noNO3, the difference in two simulations between M2000 and M2000_noNO3, and the difference in two simulations between M2010 and M2010_noNO3).

India, but the magnitudes over three regions are all relatively small due to low emissions of NO_x during this time period. The magnitudes of cooling in the industrial regions of these four areas are close to each other. However, because of the relatively less polluted western United States and northern Europe, area averaged RE_{ari} is slightly lower in U.S. and Europe compared to eastern China and India in 1975 as shown in Figure 7.

In 2000 and 2010, due to stricter regulation and introduction of new technology to on-road vehicles and fuel combustion NO_x emissions in U.S. and Europe (i.e., https://www.eea.europa.eu/data-and-maps/indicators/eea31-nox-emissions and https://www.epa.gov/air-emissions-inventories/air-pollutant-emissions-trends-data), we find that NO_x emissions in these two regions quickly drop (see Figures 3 and 5 in Hoesly et al. (2018)). Compared to 1975, the magnitude of RE_{ari} of nitrate in U.S. slightly increase from −0.047 W m^{-2} to −0.049 W m^{-2}. The magnitude of RE_{ari} of nitrate in Europe slightly increases from −0.067 to −0.072 W m^{-2}. These slight increases are probably due to enhanced NH_3 emissions and reduced SO_2 emissions, which favors more nitrate aerosol formation. Because of the rapid development in China and India during this time period, the cooling effects associated with nitrate RE_{ari} are much stronger. From 1975 to 2010, the anthropogenic emissions of NO_x in China and India jumped from 5.5% and 2.6% of total global emissions to 23.6% and 9.4%, respectively; meanwhile, the global emissions of anthropogenic NO_x increased by 58.7% in this period–meaning that the anthropogenic emissions of NO_x in China and India increased by 5.8 times and 4.7 times, respectively. (Meanwhile global NO_x emissions from natural sources increases by 8%.)

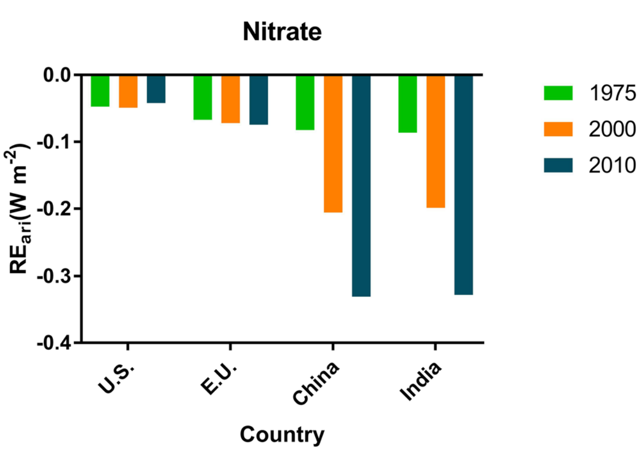

Figure 7. RE_{ari} associated with nitrate over U.S., Europe, China, and India in 1975, 2000, and 2010 periods. (U.S.: only contiguous United States is considered. Europe: defined as the European land in 35°N–70°N and 25°W–60°E. China: only eastern part of China (western boundary: 105°E) is considered. The whole India is considered).

In 2000, the cooling effect associated with nitrate RE_{ari} over eastern China reaches −0.205 W m^{-2}. In heavily polluted regions, such as Jing-Jin-Ji region and Sichuan basin, the magnitude of cooling reaches as high as −1 W m^{-2}. In the same period, we see relatively large significant cooling (−0.198 W m^{-2}) in India, which is mainly contributed by the heavy pollution in the Indo-Gangetic plain, with a magnitude about −1 W m^{-2}. In 2010, the RE_{ari} associated with nitrate in China further increases in magnitude to −0.282 W m^{-2}, which is largely contributed by the increased nitrate burden along the Southern China coast. In India, the magnitude of RE_{ari} increases tremendously to −0.331 W m^{-2}. The RE_{ari} over the entire eastern China is over −0.1 W m^{-2}, while the magnitude of RE_{ari} over the Indo-Gangetic plain is still large, at about −1 W m^{-2}. The global mean RE_{ari} of nitrate aerosols changes from −0.035 W m^{-2} in 2000 to −0.04 W m^{-2} in 2010.

Relative to 1975, the magnitude of global mean RE_{ari} of nitrate increases by 33.3% in 2010, while the values in India and eastern China increase by 2.8 and 3 times, respectively. For the purpose of comparison, we also include the RE_{ari} of sulfate aerosols and BC aerosols for the same period in Figure S2. Global mean RE_{ari} of sulfate almost remains the same from 1975 to 2010-value decreases by 2% only, from −0.244 W m^{-2} to −0.239 W

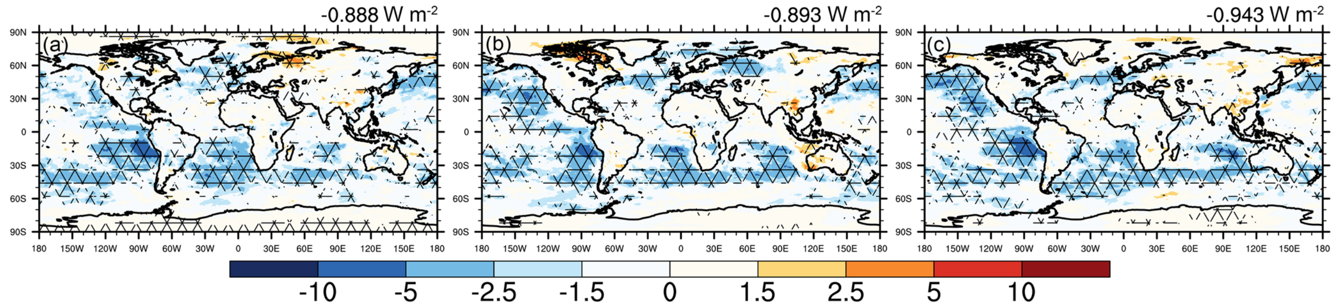

Figure 8. RE_{aci} of nitrate (unit: W m^{-2}) estimated from three pairs of experiments (from left to right: the difference in two simulations between M1975 and M1975_noNO3, the difference in two simulations between M2000 and M2000_noNO3, and the difference in two simulations between M2010 and M2010_ noNO3). Areas with meshed grids indicate the values with significance level passing 5%.

m^{-2}, but with quite different spatial patterns of change. For example, the RE_{ari} of sulfate in U.S. and Europe decreased by 52.1% and 66.4% from 1975 to 2010, while the values in India and eastern China increased by 2.2 and 1.5 times, respectively, as shown in Figure S3. Note that the RE_{ari} of BC shown in Figures S2 and S3 include not only anthropogenic but also natural sources, such as biomass burning aerosols emitted from wildfires. The RE_{ari} of black carbon in Europe drops by 8.2% from 1975 to 2000 but bounce back to be 9.7% higher in 2010 compared to 1975; while the values in eastern China and India increase by 1.41 and 1.30 times, respectively. The comparison highlights the rapidly growing importance of nitrate aerosols, especially in developing countries.

Compared to RE_{ari}, RE_{aci} signal is noisier because of complicated feedback processes among aerosols, clouds, and thermodynamic fields. By averaging 20 years' climatology simulations, we are able to get a clear picture of RE_{aci} of nitrate aerosols as shown in Figure 8. The triangle meshes in Figure 8 represent that RE_{aci} signals pass 5% significance level. The global mean RE_{aci} of nitrate in 1975, 2000, 2010 are −0.888 W m^{-2}, −0.893 W m^{-2}, and −0.943 W m^{-2}, respectively. The changes in RE_{aci} from 1975 to 2010 is 5.5 times of the changes in RE_{ari} (−0.055 W m^{-2} vs. −0.01 Wm^{-2}). These large cooling effects seem counterintuitive–for example, in Xu and Penner (2012), the first indirect effect of nitrate is −0.23 W m^{-2} in their present-day simulation. To aid in the interpretation, we further examine the changes in cloud fields, including liquid water path (ΔLWP), cloud droplet number concentration (ΔCDNC), and low-level cloud fraction (ΔCF), as well as surface temperature (ΔT_s) between cases with and without nitrate in Figure 9.

First, because of biomass burning, soil, and lightning sources of NO_x in the southern hemisphere, nitrate can enhance CCN activity as shown in Figure S4 and exerts a strong cooling effect via altering the CDNC of semi-permanent subtropical stratocumulus cloud decks as shown in Figure 9. Over southeastern Pacific along the coast of South America, the magnitude of cooling effect can be as high as 5 W m^{-2}. To analyze the source of CCN, we examine the relative difference in cloud-borne aerosol number concentrations in Figure S5. Apparently, higher CDNCs in the southern hemisphere subtropical regions are due to more than 20% of increase in cloud-borne accumulation mode aerosol number concentration. This seems counterintuitive since as shown in Figure 5, the accumulation mode nitrate aerosol burden in the southern hemisphere is low. This is because the following two facts. First, large amount of accumulation mode nitrate aerosols is first formed through aqueous chemistry in cloud-borne state then resuspend to the interstitial state. Cloud-borne state aerosols only take very small amount of total aerosol burden, like <1%. In addition, cloud-borne aerosols can be quickly depleted through wet-removal process before returning to the interstitial state, especially in the southern hemisphere subtropical regions. Second, nitrate aerosols with high hygroscopicity can sufficiently enhance CCN activity of accumulation mode aerosols in these regions. In the northern hemisphere, we can see the cooling effects of nitrate aerosols over western Pacific, eastern Pacific, and northern Atlantic, which are likely associated with the plume outflows of nitrate emitted in China, United States, and Europe, respectively.

Compared to Xu and Penner (2012), we also consider the adjustment of cloud fields to the aerosol indirect effect. For example, enhanced LWP over oceans collocates with higher CDNC, indicating more cloudiness and possible longer cloud lifetime caused by nitrate aerosols. The CF adjustment is a little more complicated

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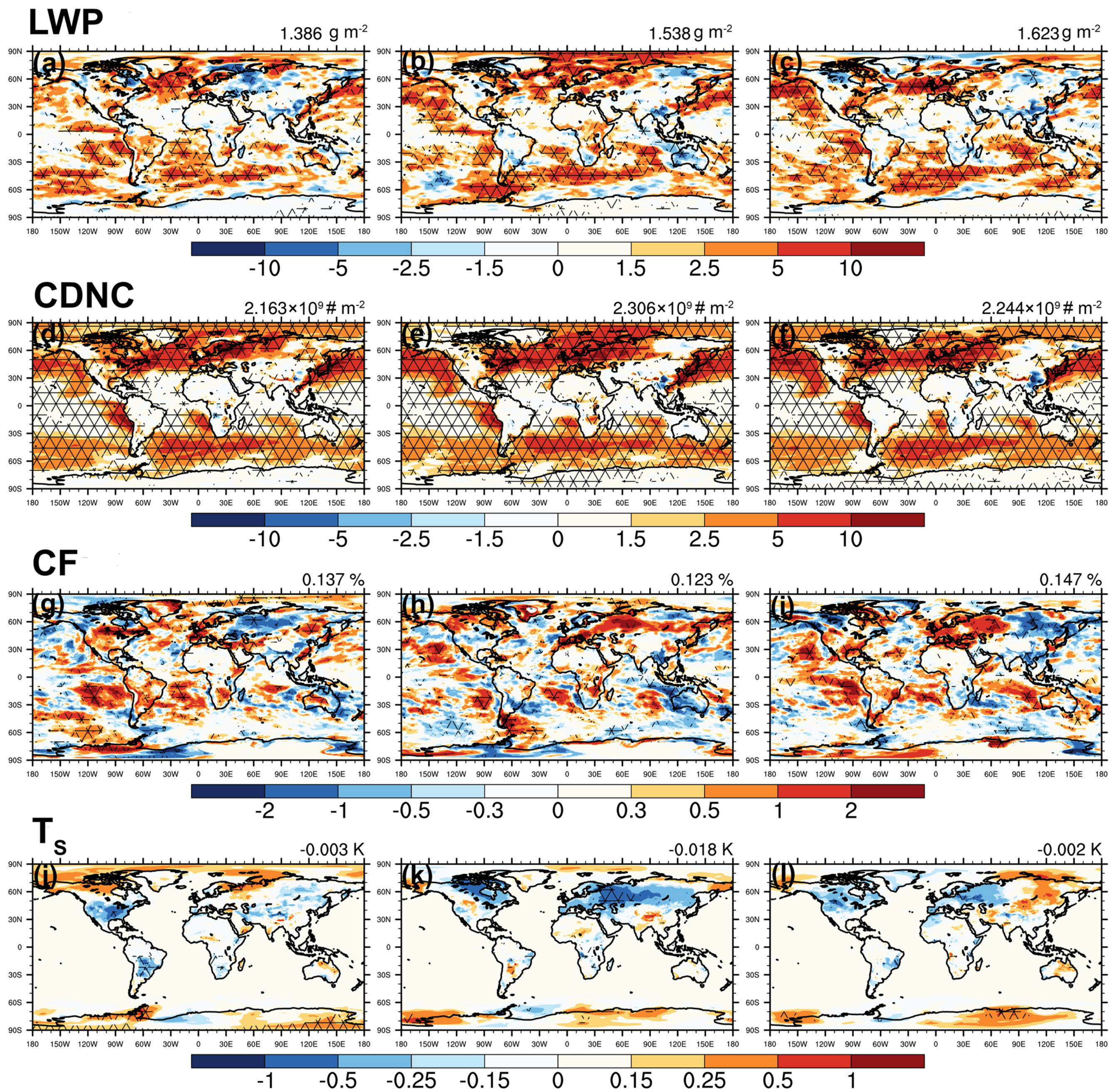

Figure 9. Differences in liquid water path (LWP), cloud droplet number concentration (CDNC), low level cloud fraction (CF), and surface temperature (T_s) from three pairs of experiments (from left to right: the difference in two simulations between M1975 and M1975_noNO3, the difference in two simulations between M2000 and M2000_noNO3, and the difference in two simulations between M2010 and M2010_noNO3). The units are shown in the upper right of each figures. Areas with meshed grids indicate the values with significance level passing 5%.

compared to LWP adjustment since it is not treated as a cloud microphysical response in the model but rather estimated in the cloud macrophysical scheme. Less regions with signals passing 5% significance level indicate large variance in the ΔCF fields.

Over land, RE_{aci} are very small, in terms of absolute values and significance levels. This agrees with smaller increase in cloud-borne aerosol number concentrations over land than ocean as shown in Figure S5. This is because based on our study, large amount of nitrate is formed through gaseous HNO_3 neutralizing NH_3 (i.e., $HNO_3 + NH_3 \leftrightarrow NH_4NO_3$) other than aqueous-phase chemistry over land. It is not surprising to see a positive RE_{aci} of nitrate aerosols in high latitude regions, since through interacting with longwave radiation, more low-level clouds due to nitrate tend to warm the environment there. However, interestingly, we find some positive signals in RE_{aci} in China, which correspond to the decreases in CDNC. At first, since we show grid-averaged CDNC in Figure 9, we thought this is due to reduced cloudiness. However, when we examine in-cloud only CDNC field as shown in Figure S4, CDNC is still reduced in China. We believe that the reduction of CDNC is due to the competition between nitrate aerosols in the coarse mode and sulfate in the accumulation mode as CCN. Ghan et al. (1998) studied a similar case, in which inclusion of more sea salt aerosols can reduce total activated cloud droplets if pre-existing sulfate loading is high (e.g., 1000 cm^{-3}) and updraft is weak. This is due to a reduction in maximum supersaturation in clouds. As noted in (Ghan et al., 1998), sea salt and sulfate must be externally mixed–if internally mixed, under any circumstance, more cloud droplets will be activated.

In our study, the accumulation-mode sulfate concentrations in China have been always extremely high since 1975, whereas the coarse-mode nitrate concentrations in China substantially grow from 1975 to 2010. Higher fraction of nitrate aerosols in the coarse mode sufficiently enhances this mode's hygroscopicity. As a result, the aerosols in two modes compete with each other for water vapor and therefore reduce the maximum supersaturation (not shown). Whenever updraft in clouds is weak, the supersaturation cannot sustain the activation of all particles in both modes as well as the growth of cloud droplets. This is more obvious in 2000 and 2010 when both sulfate and nitrate concentrations are very high. It does not occur in India's Indo-Gangetic plain probably because of the strong updraft in clouds during pre-monsoon and monsoon season.

As shown in Figure 9, surface temperature changes ΔT_s only appear over land, because SST values are prescribed in these historical climatology simulations. Any changes in cloud fields due to the inclusion of nitrate aerosols can feedback to the thermodynamic fields including T_s. For instance, if cloudiness is enhanced due to nitrate aerosols, T_s will decrease due to less incoming solar radiation. When examining ΔLWP and ΔT_s fields together, we find that positive ΔLWP is usually corresponding to negative ΔT_s, and vice versa, indicating that microphysical changes in clouds are the driving force. Occasionally, there are also some counterexamples–for instance, in Australia in M2010 versus M2010_noNO3 cases.

3.5. Radiative Forcings of Nitrate Aerosols Since 1850

Figure 10 shows the RF of nitrate aerosols, which is calculated as the difference between RE of year 2010 and RE of year 1850. Not surprisingly high RF_{ari} values locate in eastern China, India especially Indo-Gangetic plain, Europe and eastern U.S. Cooling signals over other regions, like Southern Africa and South America are likely due to difference in biomass burning emissions between 2010 and 1850. The global mean RF_{ari} of nitrate aerosols is −0.014 W m^{-2}, which is lower than the value range (−0.12 W m^{-2} to −0.02 W m^{-2}) of AeroCom Phase II models as reported in Myhre, Samset, et al. (2013), and many other studies (e.g., −0.14 W m^{-2} in An et al. (2019); −0.025 W m^{-2} in Li et al. (2015); −0.056 W m^{-2} in Hauglustaine et al. (2014); −0.14 W m^{-2} in Xu and Penner (2012)).

The main reason for our smaller magnitude of RF_{ari} is due to smaller nitrate burden (as shown in Table 3) and lower mass fraction of fine particles (accumulation + Aitken modes) in the total nitrate burden compared to previous studies. According to Hauglustaine et al. (2014), fine particles contribute to about 90% of forcing because of their high number concentrations. The fine mode mass fraction of nitrate in our study is 28%, which is exactly the same as in Hauglustaine et al. (2014), but lower than many other models/studies (e.g., 52.29% in Xu and Penner (2012), and many AeroCom models in Bian et al. (2017)). Lower fine mode mass fraction is due to the fact that, in gas-aerosol exchange process, more nitrate aerosol mass tends to partition on the coarse mode eventually. This process is further exacerbated due to small σ (standard deviation of lognormal distribution) of the coarse mode in CESM2 ($\sigma = 1.2$ in CESM2 while $\sigma = 1.8$ in CESM1), which leads to a longer lifetime of coarse mode aerosols in the atmosphere.

Second, the negative RF_{ari} of nitrate aerosols over some biomass burning regions, like Indonesia and the Sahel region, is not as strong as in previous studies (e.g., An et al., 2019; Hauglustaine et al., 2014). This difference is likely due to the difference in CEDS and CMIP5 emission datasets. In addition, we predicted small but positive RF_{ari} of nitrate aerosols over vast regions, including high latitudes, Southern Ocean, and

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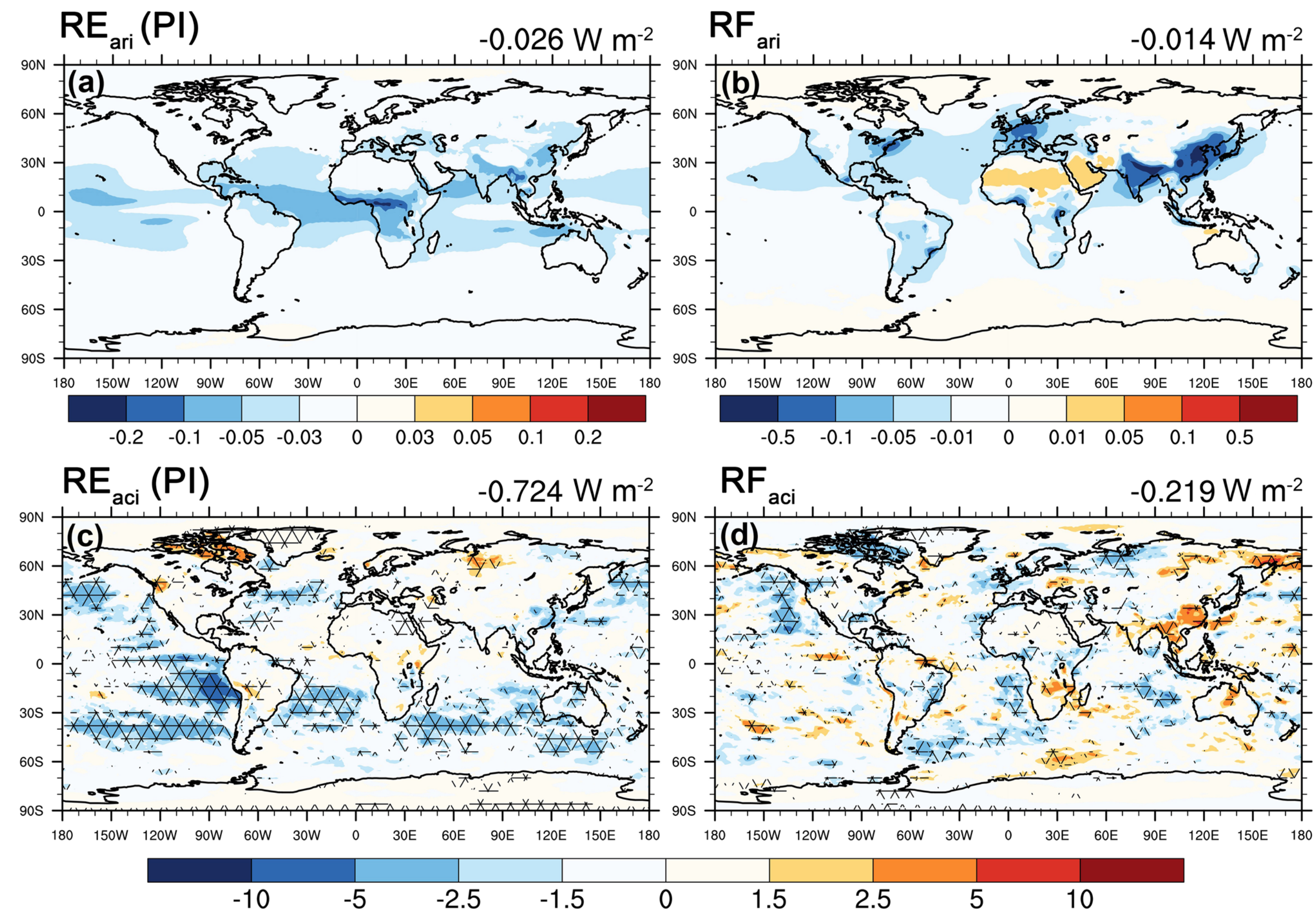

Figure 10. RE$_{ari}$ of nitrate aerosols in 1850 and RF$_{ari}$ of nitrate aerosols since 1850 (upper panel); RE$_{aci}$ of nitrate aerosols in 1850 and RF$_{aci}$ of nitrate aerosols since 1850 (lower panel).

dust source regions. Warming effect can be seen over high latitude regions because of bright surface with high albedo. Such a warming effect is similar to the GISS-MATRIX and INCA model simulations in Myhre, Samset, et al. (2013) and An et al. (2019). Over the Southern Ocean where sea salt dominates aerosol populations, including nitrate aerosols on sea salt actually reduces the water uptake ability of sea salt aerosols, which then decreases the extinction and leads to a warming effect. This effect can be seen in Xu and Penner (2012). In the dust source regions of Sahara Desert and Arabian Peninsula, including nitrate aerosols enhance the water uptake ability of dust aerosols, which increases longwave emissivity of dust aerosols. As a result, the longwave warming effect of dust aerosols becomes stronger. Some AeroCom Phase II models (likely OsloCTM2 and NCAR-CAM3.5) also produced such a warming effect (Myhre, Samset, et al., 2013). In our study, the RF$_{ari}$ of nitrate aerosols is around 0.01 W m^{-2} to 0.03 W m^{-2} in these regions.

As shown in Figure 10, the RF$_{aci}$ pattern looks noisy. The global mean RF$_{aci}$ is −0.219 W m^{-2}, which is larger than the values reported by two previous studies that considered the first aerosol indirect effect only (e.g., −0.09 W m^{-2} in Xu and Penner (2012) and −0.05 W m^{-2} in Bellouin et al. (2011)). Reason for such a large value have been explained in the previous section. In the southern hemisphere, strong cooling signals of nitrate RE$_{aci}$ as shown in Figure 8 are mostly canceled out over the oceans. One remaining cooling of RF$_{aci}$ over southeastern Atlantic stratocumulus cloud decks is due to higher biomass burning NO_x emissions in 2010 than those in 1850. In northern hemisphere, significant cooling signals can be seen over eastern Pacific, eastern U.S., northern Atlantic, and Europe. Strong warming of RF$_{aci}$ can be seen in China because of positive RE$_{aci}$ in China in 2010 as shown in Figure 8 and negative RE$_{aci}$ in China in PI as shown in Figure 10. Besides China, there are many regions with positive RF$_{aci}$, including Southern Africa, Mainland Southeast

Asia, and part of Siberia. This is because cloud albedo is more susceptible to aerosols (nitrate aerosols in this case) in the PI period when the environment is very clean (Carslaw et al., 2013).

4. Discussion and Conclusions

In this study, we incorporated the MOSAIC module in CESM2(CAM6-chem). The module is used to simulate the dynamic gas-aerosol exchange based on several novel approaches (Zaveri et al., 2008), and the remaining aerosol processes are still treated by the default MAM4 aerosol module. One unique capability of MOSAIC is that it can efficiently simulate the gas-aerosol exchange of nitric acid gas to form nitrate aerosols. By comparing against the ground-based observations of aerosol mass concentrations (e.g., IMPROVE over U.S., EANET over East Asia, and CAWNET over China), as well as MODIS and MISR AOD products, we found that the model performs reasonably well in simulating nitrate aerosols and total aerosol loadings. By including nitrate aerosols, the mass burdens of other aerosol species are all affected to some extent, especially ammonium and sea salt aerosols.

One of the main goals of this study is to examine the RE of nitrate aerosols, especially its dramatic changes since year 1975. From 1975 to 2010, greater economic activities enhance the RE_{ari} of nitrate aerosols over eastern China, and India. In contrast, stricter regulations on emissions of NO_x (Hoesly et al., 2018) were able to bring RE_{ari} of nitrate slightly down in U.S. or only slightly increase RE_{ari} in Europe from the 1975 period to the 2010 period. Although the global mean RE_{ari} of nitrate is quite small (e.g., −0.04 W m^{-2} in 2010), the regional RE_{ari} of nitrate over heavily polluted regions, such as China and India, is still high (about −0.33 W m^{-2}), and increased rapidly–faster than any other important aerosols such as sulfate and BC (e.g., 2.8–3 times increase from 1975 to 2010).

In our study, large amounts of nitrate aerosols are formed in cloud-borne state accumulation mode aerosols through aqueous chemistry (e.g., uptake of NH_3 and HNO_3 in cloud droplets), then resuspend to interstitial state aerosols, and meanwhile the accumulation mode nitrate aerosols are losing mass to the coarse mode nitrate aerosols via gas-aerosol exchange process. This whole process is extremely important in our study, because (a) it leads to small fine particle mass fraction, which then further leads to small RE_{ari}/RF_{ari} of nitrate aerosols (0.04 W m^{-2}/0.014 W m^{-2}); and (b) it significantly increases the cloud-borne state accumulation mode aerosols (e.g., by 20%) in semi-permanent cloud decks in southern hemisphere and causes a large cooling of RE_{aci}. Larger RE_{aci}/RF_{aci} (about −0.9 W m^{-2}/−0.22 W m^{-2}) compared to previous studies (Bellouin et al., 2011; Xu & Penner, 2012) is also due to the fact that we considered cloud adjustment to nitrate aerosols (e.g., enhanced LWP) in addition to the first aerosol indirect effect. Low RE_{ari}/RF_{ari} but high RE_{aci}/RF_{aci} is likely partially due to highly sensitive aerosol-cloud interaction in CESM2 (Gettelman, et al., 2019). In China, competition between high sulfate loading in the accumulation mode and high nitrate loading in the coarse mode reduce the maximum supersaturation, reduce CDNC, and exert a warming effect, especially in 2010, which results in strong warming RF_{aci} in China.

In summary, the total RF caused by nitrate aerosols is −0.233 W m^{-2}, which is quite significant if compared to other RF agents shown in the IPCC AR5 report. From 1975 to 2010, with 58.7% more NO_x anthropogenic emission, the total RE increases in magnitude by −0.065 W m^{-2}. This highlights the importance of inclusion of nitrate aerosols in any climate simulations.

Data Availability Statement

NASA MODIS and MISR datasets are downloaded from https://earthdata.nasa.gov/. IMPROVE data set is downloaded from http://vista.cira.colostate.edu/Improve/. EANET data set is downloaded from https://monitoring.eanet.asia/document/public/index. NASA INTEX-B data set (used in the supplementary) is downloaded from https://www-air.larc.nasa.gov/missions/intex-b. Model CESM2 can be find at: https://www.cesm.ucar.edu/models/cesm2/following downloading instructions. Outputs generated for this study can be accessed: https://zenodo.org/record/4979841#.YMw6AWg3mUl.

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References

Aan de Brugh, J., Schaap, M., Vignati, E., Dentener, F., Kahnert, M., Sofiev, M., et al. (2011). The European aerosol budget in 2006. *Atmospheric Chemistry and Physics*, *11*, 1117–1139. https://doi.org/10.5194/acp-11-111-2011

Abdul-Razzak, H., & Ghan, S. J. (2000). A parameterization of aerosol activation: 2. Multiple aerosol type. *Journal of Geophysical Research*, *105*, 6837–6844. https://doi.org/10.1029/1999JD901161

Adams, P. J., Seinfeld, J. H., Koch, D., Mickley, L., & Jacob, D. (2001). General circulation model assessment of direct radiative forcing by the sulfate-nitrate-ammonium-water inorganic aerosol system. *Journal of Geophysical Research*, *106*, 1097–1111. https://doi.org/10.1029/2000JD900512

An, Q., Zhang, H., Wang, Z., Liu, Y., Xie, B., Liu, Q., et al. (2019). The development of an atmospheric aerosol/chemistry-climate model, BCC_AGCM_CUACE2.0, and simulated effective radiative forcing of nitrate aerosols. *Journal of Advances in Modeling Earth Systems*, *11*, 3816–3835. https://doi.org/10.1029/2019MS001622

Bauer, S. E., Koch, D., Unger, N., Metzger, S. M., Shindell, D. T., & Streets, D. G. (2007). Nitrate aerosols today and in 2030: A global simulation including aerosols and tropospheric ozone. *Atmospheric Chemistry and Physics*, *7*, 5043–5059. https://doi.org/10.5194/acp-7-5043-2007

Bellouin, N., Rae, J., Jones, A., Johnson, C., Haywood, J., & Boucher, O. (2011). Aerosol forcing in the Climate Model Intercomparison Project (CMIP5) simulations by HadGEM2-ES and the role of ammonium nitrate. *Journal of Geophysical Research*, *116*, D20206. https://doi.org/10.1029/2011JD016074

Bian, H., Chin, M., Hauglustaine, D. A., Schulz, M., Myhre, G., Bauer, S. E., et al. (2017). Investigation of global particulate nitrate from the AeroCom phase III experiment. *Atmospheric Chemistry and Physics*, *17*, 12911–12940. https://doi.org/10.5194/acp-17-12911-2017

Bogenschutz, P. A., Gettelman, A., Hannay, C., Larson, V. E., Neale, R. B., Craig, C., & Chen, C. C. (2018). The path to CAM6: Coupled simulations with CAM5.4 and CAM5.5. *Geoscientific Model Development*, *11*, 1–38. https://doi.org/10.5194/gmd-2017-129

Carslaw, K. S., Lee, L. A., Reddington, C. L., Pringle, K. J., Rap, A., Forster, P. M., et al. (2013). Large contribution of natural aerosols to uncertainty in indirect forcing. *Nature*, *503*(7474), 67–71. https://doi.org/10.1038/nature12674

Danabasoglu, G., Lamarque, J. F., Bacmeister, J., Bailey, D. A., DuVivier, A. K., Edwards, J., et al. (2020). The Community Earth System Model Version 2 (CESM2). *Journal of Advances in Modeling Earth Systems*, *12*(2). https://doi.org/10.1029/2019MS001916

Emmons, L. K., Schwantes, R. H., Orlando, J. J., Tyndall, G., Kinnison, D., Lamarque, J.-F., et al. (2020). The Chemistry Mechanism in the Community Earth System Model version 2 (CESM2). *Journal of Advances in Modeling Earth Systems*, *12*, e2019MS001882. https://doi.org/10.1029/2019MS001882

Fan, T., Liu, X., Ma, P. L., Zhang, Q., Li, Z., Jiang, Y., et al. (2018). Emission or atmospheric processes? An attempt to attribute the source of large bias of aerosols in eastern China simulated by global climate models. *Atmospheric Chemistry and Physics*, *18*(2), 1395–1417. https://doi.org/10.5194/acp-18-1395-2018

Fast, J. D., Gustafson, W. I., Easter, R. C., Zaveri, R. A., Barnard, J. C., Chapman, E. G., et al. (2006). Evolution of ozone, particulates, and aerosol direct radiative forcing in the vicinity of Houston using a fully coupled meteorology-chemistry-aerosol model. *Journal of Geophysical Research*, *111*, D21305. https://doi.org/10.1029/2005JD006721

Feng, Y., & Penner, J. E. (2007). Global modeling of nitrate and ammonium: Interaction of aerosols and tropospheric chemistry. *Journal of Geophysical Research*, *112*, D01304. https://doi.org/10.1029/2005JD006404

Gao, Y., Liu, X., Zhao, C., & Zhang, M. (2011). Emission controls versus meteorological conditions in determining aerosol concentrations in Beijing during the 2008 Olympic Games. *Atmospheric Chemistry and Physics*, *11*(23), 12–12. https://doi.org/10.5194/acp-11-12437-2011437451

Gettelman, A., Hannay, C., Bacmeister, J. T., Neale, R. B., Pendergrass, A. G., Danabasoglu, G., et al. (2019). High climate sensitivity in the Community Earth System Model Version 2 (CESM2). *Geophysical Research Letters*, *46*, 8329–8337. https://doi.org/10.1029/2019GL083978

Gettelman, A., Morrison, H., Santos, S., Bogenschutz, P., & Caldwell, P. M. (2015). Advanced two-moment bulk microphysics for global models. Part II: Global model solutions and aerosol-cloud interactions. *Journal of Climate*, *28*(3), 1288–1307. https://doi.org/10.1175/jcli-d-14-00103.1

Ghan, S. J. (2013). Technical note: Estimating aerosol effects on cloud radiative forcing. *Atmospheric Chemistry and Physics*, *13*(19), 9971–9974. https://doi.org/10.5194/acp-13-9971-2013

Ghan, S. J., Guzman, G., & Abdul-Razzak, H. (1998). Competition between sea salt and sulfate particles as cloud condensation nuclei. *Journal of the Atmospheric Sciences*, *55*(22), 3340–3347. https://doi.org/10.1175/1520-0469(1998)055<3340:cbssas>2.0.co;2

Golaz, J. -C., Larson, V. E., & Cotton, W. R. (2002). A pdf-based model for boundary layer clouds: Part I. Method and model description. *Journal of the Atmospheric Sciences*, *59*(24), 3540–3551. https://doi.org/10.1175/1520-0469(2002)059<3540:apbmfb>2.0.co;2

Hauglustaine, D. A., Balkanski, Y., & Schulz, M. (2014). A global model simulation of present and future nitrate aerosols and their direct radiative forcing of climate. *Atmospheric Chemistry and Physics*, *14*, 11031–11063. https://doi.org/10.5194/acp-14-11031-2014

Hoesly, R. M., Smith, S. J., Feng, L., Klimont, Z., Janssens-Maenhout, G., Pitkanen, T., et al. (2018). Historical (1750-2014) anthropogenic emissions of reactive gases and aerosols from the Community Emissions Data System (CEDS). *Geoscientific Model Development*, *11*, 369–408. https://doi.org/10.5194/gmd-11-369-2018

Huang, X., Ding, A., Gao, J., Zheng, B., Zhou, D., Qi, X., et al. (2020). Enhanced secondary pollution offset reduction of primary emissions during COVID-19 lockdown in China. *National Science Review*, *8*. https://doi.org/10.1093/nsr/nwaa137

Hurrell, J. W., Holland, M. M., Gent, P. R., Ghan, S., Kay, J. E., Kushner, P. J., et al. (2013). The community earth system model: A framework for collaborative research. *Bulletin of the American Meteorological Society*, *94*, 1339–1360. https://doi.org/10.1175/bams-d-12-00121.1

Iacono, M. J., Delamare, J. S., Mlawer, E. J., Shephard, M. W., Clough, S. A., & Collins, W. D. (2008). Radiative forcing by long-lived greenhouse gases: Calculations with the AER radiative transfer models. *Journal of Geophysical Research*, *113*, D13103. https://doi.org/10.1029/2008JD0099410.1029/2008jd009944

Kahn, R. A., Garay, M. J., Nelson, D. L., Yau, K. K., Bull, M. A., Gaitley, B. J., et al. (2007). Satellite-derived aerosol optical depth over dark water from MISR and MODIS: Comparisons with AERONET and implications for climatological studies. *Journal of Geophysical Research*, *112*, D18205. https://doi.org/10.1029/2006JD008175

Kang, N., Kumar, K. R., Hu, K., Yu, X., & Yin, Y. (2016). Long-term (2002–2014) evolution and trend in Collection 5.1 Level-2 aerosol products derived from the MODIS and MISR sensors over the Chinese Yangtze River Delta. *Atmospheric Research*, *181*, 29–43. https://doi.org/10.1016/j.atmosres.2016.06.008

Lamarque, J. F., Emmons, L. K., Hess, P. G., Kinnison, D. E., Tilmes, S., Vitt, F., et al. (2012). CAM-chem: Description and evaluation of interactive atmospheric chemistry in the Community Earth System Model. *Geoscientific Model Development*, *5*(2), 369–411. https://doi.org/10.5194/gmd-5-369-2012

Larson, V. E., Golaz, J. C., & Cotton, W. R. (2002). Small-scale and mesoscale variability in cloudy boundary layers: Joint probability density functions. *Journal of the Atmospheric Sciences*, *59*, 3519–3539. https://doi.org/10.1175/1520-0469(2002)059<3519:SSAMVI>2.0.CO;2

Levy, R. C., Remer, L. A., Kleidman, R. G., Mattoo, S., Ichoku, C., Kahn, R., & Eck, T. F. (2010). Global evaluation of the Collection 5 MODIS dark-target aerosol products over land. *Atmospheric Chemistry and Physics*, *10*, 10399–10420. https://doi.org/10.5194/acp-10-10399-2010

Li, J., Wang, W., Liao, H., & Chang, W. (2015). Past and future direct radiative forcing of nitrate aerosol in East Asia. *Theoretical and Applied Climatology*, *121*, 445–458. https://doi.org/10.1007/s00704-014-1249-1

Liao, H., Seinfeld, J. H., Adams, P. J., & Mickley, L. J. (2004). Global radiative forcing of coupled tropospheric ozone and aerosols in a unified general circulation model. *Journal of Geophysical Research*, *109*, D16207. https://doi.org/10.1029/2003JD004456

Liu, X., Easter, R. C., Ghan, S. J., Zaveri, R., Rasch, P., Shi, X., et al. (2012). Toward a minimal representation of aerosols in climate models: Description and evaluation in the Community Atmosphere Model CAM5. *Geoscientific Model Development*, *5*, 709–739. https://doi.org/10.5194/gmd-5-709-2012

Liu, X., Ma, P. L., Wang, H., Tilmes, S., Singh, B., Easter, R. C., et al. (2016). Description and evaluation of a new four-mode version of the modal aerosol module (MAM4) within version 5.3 of the Community Atmosphere Model. *Geoscientific Model Development*, *9*(2), 505–522. https://doi.org/10.5194/gmd-9-505-2016

Liu, X., & Penner, J. E. (2005). Ice nucleation parameterization for global models. *Meteorologische Zeitschrift*, *14*. https://doi.org/10.1127/0941-2948/2005/0059

Liu, X., Penner, J. E., Ghan, S. J., & Wang, M. (2007). Inclusion of ice microphysics in the NCAR Community Atmosphere Model version 3 (CAM3). *Journal of Climate*, *20*, 4526–4547. https://doi.org/10.1175/jcli4264.1

Lu, Z., Zhang, Q., & Streets, D. G. (2011). Sulfur dioxide and primary carbonaceous aerosol emissions in China and India, 1996–2010. *Atmospheric Chemistry and Physics*, *11*, 9839–9864. https://doi.org/10.5194/acp-11-9839-2011

Malm, W. C., Schichtel, B. A., Pitchford, M. L., Ashbaugh, L. L., & Eldred, R. A. (2004). Spatial and monthly trends in speciated fine particle concentration in the United States. *Journal of Geophysical Research*, *109*, D03306. https://doi.org/10.1029/2003jd003739

Morrison, H., & Gettelman, A. (2008). A new two-moment bulk stratiform cloud microphysics scheme in the Community Atmosphere Model, version 3(CAM3): Part I. Description and numerical tests. *Journal of Climate*, *21*, 3642–3659. https://doi.org/10.1175/2008JCLI2105.1

Myhre, G., Samset, B. H., Schulz, M., Balkanski, Y., Bauer, S., Berntsen, T. K., et al. (2013). Radiative forcing of the direct aerosol effect from AeroCom Phase II simulations. *Atmospheric Chemistry and Physics*, *13*, 1853–1877. https://doi.org/10.5194/acp-13-1853-2013

Myhre, G., Shindell, D., Bréon, F.-M., Collins, W., Fuglestvedt, J., Huang, J., et al. (2013). Anthropogenic and Natural Radiative Forcing. In T. F. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S. K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex, & P. M. Midgley (Eds.), *Climate change 2013: The physical science basis. Contribution of working group I to the fifth assessment report of the intergovernmental panel on climate change*. Cambridge University Press.

Nenes, A., Pandis, S. N., & Pilinis, C. (1998). ISORROPIA: A new thermodynamic equilibrium model for multiphase multicomponent inorganic aerosols. *Aquatic Geochemistry*, *4*(1), 123–152. https://doi.org/10.1023/a:1009604003981

Pagalan, L., Winters, M., Lanphear, B., Brauer, M., Bickford, C., Lanphear, N., et al. (2018). Associations between Prenatal Exposure to Air Pollutants and Developing Autism: A Population-Based Cohort Study in Metro Vancouver, Canada. In *ISEE conference abstracts*. https://doi.org/10.1289/isee.2017.2017-475

Petters, M., & Kreidenweis, S. (2007). A single parameter representation of hygroscopic growth and cloud condensation nucleus activity. *Atmospheric Chemistry and Physics*, *7*(8), 1961–1971. https://doi.org/10.5194/acp-7-1961-2007

Pöschl, U. (2005). Atmospheric aerosols: Composition, transformation, climate and health effects. *Angewandte Chemie International Edition*, *44*, 7520–7540. https://doi.org/10.1002/anie.200501122

Rodriguez, M. A., & Dabdub, D. (2004). IMAGES-SCAPE2: A modeling study of size- and chemically resolved aerosol thermodynamics in a global chemical transport model. *Journal of Geophysical Research*, *109*, D02203. https://doi.org/10.1029/2003JD003639

Shi, X., Liu, X., & Zhang, K. (2015). Effects of pre-existing ice crystals on cirrus clouds and comparison between different ice nucleation parameterizations with the Community Atmosphere Model (CAM5). *Atmospheric Chemistry and Physics*, *15*(3), 1503–1520. https://doi.org/10.5194/acp-15-1503-2015

Skeie, R. B., Berntsen, T. K., Myhre, G., Tanaka, K., Kvalevåg, M. M., & Hoyle, C. R. (2011). Anthropogenic radiative forcing time series from pre-industrial times until 2010. *Atmospheric Chemistry and Physics*, *11*(22), 11827–11857. https://doi.org/10.5194/acp-11-11827-2011

Smith, S. J., van Aardenne, J., Klimont, Z., Andres, R. J., Volke, A., & Arias, S. D. (2011). Anthropogenic sulfur dioxide emissions: 1850–2005. *Atmospheric Chemistry and Physics*, *11*, 1101–1116. https://doi.org/10.5194/acp-11-1101-2011

van Marle, M., Kloster, S., Magi, B., Marlon, J., Daniau, A., Field, R., et al. (2017). Historic global biomass burning emissions for CMIP6 (BB4CMIP) based on merging satellite observations with proxies and fire models (1750-2015). *Geoscientific Model Development*, *10*(9), 3329–3357. https://doi.org/10.5194/gmd-10-3329-2017

Wang, G., Zhang, R., Gomez, M. E., Yang, L., Zamora, M. L., Hu, M., et al. (2016). Persistent sulfate formation from London Fog to Chinese haze. *Proceedings of the National Academy of Sciences*, *113*(48), 13630–13635. https://doi.org/10.1073/pnas.1617361113

Wang, Y., Liu, X., Hoose, C., & Wang, B. (2014). Different contact angle distributions for heterogeneous ice nucleation in the Community Atmospheric Model version 5. *Atmospheric Chemistry and Physics*, *14*, 10411–10430. https://doi.org/10.5194/acp-14-10411-2014

Xi, X., & Sokolik, I. N. (2015). Dust interannual variability and trend in Central Asia from 2000 to 2014 and their climatic linkages. *Journal of Geophysical Research: Atmospheres*, *120*, 12175–12197. https://doi.org/10.1002/2015JD024092

Xu, L., & Penner, J. E. (2012). Global simulations of nitrate and ammonium aerosols and their radiative effects. *Atmospheric Chemistry and Physics*, *12*, 9479–9504. https://doi.org/10.5194/acp-12-9479-2012

Yang, Q., Gustafson, W. I., Fast, J. D., Wang, H., Easter, R. C., Morrison, H., et al. (2011). Assessing regional scale predictions of aerosols, marine stratocumulus, and their interactions during VOCALS-REx using WRF-Chem. *Atmospheric Chemistry and Physics*, *11*(23), 11951–11975. https://doi.org/10.5194/acp-11-11951-2011

Zaveri, R. A., Easter, R. C., Fast, J. D., & Peters, L. K. (2008). Model for Simulating Aerosol Interactions and Chemistry (MOSAIC). *Journal of Geophysical Research*, *113*, D13204. https://doi.org/10.1029/2007JD008782

Zaveri, R. A., Easter, R. C., Singh, B., Wang, H., Lu, Z., Tilmes, S., et al. (2021). Development and evaluation of chemistry-aerosol-climate model CAM5-chem-MAM7-MOSAIC: Global atmospheric distribution and radiative effects of nitrate aerosol. *Journal of Advances in Modeling Earth Systems*, *13*, e2020MS002346. https://doi.org/10.1029/2020MS002346

Zhang, Q., Jimenez, M. R., Canagaratna, M. R., Allan, J. D., Coe, H., Ulbrich, I., et al. (2007). Ubiquity and dominance of oxygenated species in organic aerosols in anthropogenically-influenced Northern Hemi- sphere midlatitudes. *Geophysical Research Letters*, *34*, L13801. https://doi.org/10.1029/2007gl02997910.1029/2007gl029979

Zhang, X. Y., Wang, Y. Q., Zhang, X. C., Guo, W., & Gong, S. L. (2008). Carbonaceous aerosol composition over various regions of China during 2006. *Journal of Geophysical Research*, *113*, D14111. https://doi.org/10.1029/2007JD009525

Zhao, C., Tie, X., & Lin, Y. (2006). A possible positive feedback of reduction of precipitation and increase in aerosols over eastern central China. *Geophysical Research Letters*, *33*(11), L11814. https://doi.org/10.1029/2006gl025959

References From the Supporting Information

Bouwman, A. F., Lee, D. S., Asman, W. A. H., Dentener, F. J., Van Der Hoek, K. W., & Olivier, J. G. J. (1997). A global high-resolution emission inventory for ammonia. *Global Biogeochemical Cycles*, *11*, 561–587. https://doi.org/10.1029/97GB02266

Singh, H. B., Brune, W. H., Crawford, J. H., Flocke, F., & Jacob, D. J. (2009). Chemistry and transport of pollution over the Gulf of Mexico and the Pacific: Spring 2006 INTEX-B Campaign overview and first results. *Atmospheric Chemistry and Physics*, *9*, 2301–2318.

van Aardenne, J. A., Dentener, F. J., Olivier, J. G. J., Klein Goldewijk, C. G. M., & Lelieveld, J. (2001). A 1° × 1° resolution data set of historical anthropogenic trace gas emissions for the period 1890–1990. *Global Biogeochemical Cycles*, *15*(4), 909–928. https://doi.org/10.1029/2000gb001265